

STAT 140: Design of Experiments
Section 10 Solutions: Full Factorial Designs (2^K)

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Exercise 1: Contrast Vectors in a 2^3 Design

Part (a): g_{F_3}

Solution

In lexicographic order, the level of F_3 in each cell (slowest-changing = F_1 , fastest-changing = F_3) is:

$$g_{F_3} = (-1, +1, -1, +1, -1, +1, -1, +1)^T.$$

The pattern simply alternates $-1, +1$ every row because F_3 is the last (fastest-varying) factor.

Part (b): $g_{F_1 * F_3}$

Solution

We compute $g_{F_1 * F_3} = g_{F_1} \odot g_{F_3}$ entry-wise. Recall $g_{F_1} = (-1, -1, -1, -1, +1, +1, +1, +1)^T$.

Cell	$[g_{F_1}]_w$	$[g_{F_3}]_w$	$[g_{F_1 * F_3}]_w$
$(-1, -1, -1)$	-1	-1	$+1$
$(-1, -1, +1)$	-1	$+1$	-1
$(-1, +1, -1)$	-1	-1	$+1$
$(-1, +1, +1)$	-1	$+1$	-1
$(+1, -1, -1)$	$+1$	-1	-1
$(+1, -1, +1)$	$+1$	$+1$	$+1$
$(+1, +1, -1)$	$+1$	-1	-1
$(+1, +1, +1)$	$+1$	$+1$	$+1$

So $g_{F_1 * F_3} = (+1, -1, +1, -1, -1, +1, -1, +1)^T$.

Part (c): $g_{F_1 * F_2 * F_3}$ **Solution**

We compute $g_{F_1 * F_2 * F_3} = g_{F_1} \odot g_{F_2} \odot g_{F_3}$ entry-wise. Recall $g_{F_2} = (-1, -1, +1, +1, -1, -1, +1, +1)^T$.

Cell	$[g_{F_1}]_w$	$[g_{F_2}]_w$	$[g_{F_3}]_w$	$[g_{F_1 * F_2 * F_3}]_w$
$(-1, -1, -1)$	-1	-1	-1	-1
$(-1, -1, +1)$	-1	-1	+1	+1
$(-1, +1, -1)$	-1	+1	-1	+1
$(-1, +1, +1)$	-1	+1	+1	-1
$(+1, -1, -1)$	+1	-1	-1	+1
$(+1, -1, +1)$	+1	-1	+1	-1
$(+1, +1, -1)$	+1	+1	-1	-1
$(+1, +1, +1)$	+1	+1	+1	+1

So $g_{F_1 * F_2 * F_3} = (-1, +1, +1, -1, +1, -1, -1, +1)^T$.

Part (d): Explicit formula for τ_{i, F_1} **Solution**

Using $\tau_{i, F_1} = \frac{1}{4} Y_i g_{F_1}$ with $g_{F_1} = (-1, -1, -1, -1, +1, +1, +1, +1)^T$:

$$\tau_{i, F_1} = \frac{1}{4} [-Y_i(-1, -1, -1) - Y_i(-1, -1, +1) - Y_i(-1, +1, -1) - Y_i(-1, +1, +1) + Y_i(+1, -1, -1) + Y_i(+1, -1, +1) + Y_i(+1, +1, -1) + Y_i(+1, +1, +1)].$$

Verification. We can rewrite this as:

$$\tau_{i, F_1} = \frac{1}{4} [(Y_i(+1, -1, -1) - Y_i(-1, -1, -1)) + (Y_i(+1, -1, +1) - Y_i(-1, -1, +1)) + (Y_i(+1, +1, -1) - Y_i(-1, +1, -1)) + (Y_i(+1, +1, +1) - Y_i(-1, +1, +1))].$$

This is exactly the average of the effect of F_1 across all four combinations of (F_2, F_3) , i.e., averaging over $F_2 \in \{-1, +1\}$ and $F_3 \in \{-1, +1\}$, each with weight $\frac{1}{4}$. This confirms the formula correctly averages over all settings of the other factors.

Exercise 2: A 2^3 Worked Example

For reference, denote the observed group means in lexicographic order as the 1×8 row vector:

$$\bar{Y}^{obs} = (4, 6, 5, 9, 5, 7, 8, 14)$$

corresponding to cells $(-1, -1, -1)$, $(-1, -1, +1)$, $(-1, +1, -1)$, $(-1, +1, +1)$, $(+1, -1, -1)$, $(+1, -1, +1)$, $(+1, +1, -1)$, $(+1, +1, +1)$.

Part (a): Main effects $\hat{\tau}_{F_1}$, $\hat{\tau}_{F_2}$, $\hat{\tau}_{F_3}$

Solution

Using $\hat{\tau}_{F_k} = \frac{1}{4} \bar{Y}^{obs} g_{F_k}$, where $\bar{Y}^{obs}$ is the 1×8 row vector and g_{F_k} is the 8×1 column vector (row $\times$ column = scalar):

Main effect of F_1 : $g_{F_1} = (-1, -1, -1, -1, +1, +1, +1, +1)^T$.

$$\begin{aligned}\hat{\tau}_{F_1} &= \frac{1}{4} [-(4) - (6) - (5) - (9) + (5) + (7) + (8) + (14)] \\ &= \frac{1}{4} [-24 + 34] = \frac{10}{4} = 2.5.\end{aligned}$$

Main effect of F_2 : $g_{F_2} = (-1, -1, +1, +1, -1, -1, +1, +1)^T$.

$$\begin{aligned}\hat{\tau}_{F_2} &= \frac{1}{4} [-(4) - (6) + (5) + (9) - (5) - (7) + (8) + (14)] \\ &= \frac{1}{4} [-22 + 36] = \frac{14}{4} = 3.5.\end{aligned}$$

Main effect of F_3 : $g_{F_3} = (-1, +1, -1, +1, -1, +1, -1, +1)^T$.

$$\begin{aligned}\hat{\tau}_{F_3} &= \frac{1}{4} [-(4) + (6) - (5) + (9) - (5) + (7) - (8) + (14)] \\ &= \frac{1}{4} [-22 + 36] = \frac{14}{4} = 3.5.\end{aligned}$$

Part (b): Two-factor interactions**Solution**

Using the contrast vectors computed in Exercise 1 and in the tutorial notes:

$g_{F_1 * F_2} = (+1, +1, -1, -1, -1, +1, +1)^T$ (from $g_{F_1} \odot g_{F_2}$).

$$\begin{aligned}\hat{\tau}_{F_1 * F_2} &= \frac{1}{4}[(4) + (6) - (5) - (9) - (5) - (7) + (8) + (14)] \\ &= \frac{1}{4}[6] = 1.5.\end{aligned}$$

$g_{F_1 * F_3} = (+1, -1, +1, -1, -1, +1, -1, +1)^T$ (from part 1b).

$$\begin{aligned}\hat{\tau}_{F_1 * F_3} &= \frac{1}{4}[(4) - (6) + (5) - (9) - (5) + (7) - (8) + (14)] \\ &= \frac{1}{4}[2] = 0.5.\end{aligned}$$

$g_{F_2 * F_3} = g_{F_2} \odot g_{F_3} = (+1, -1, -1, +1, +1, -1, -1, +1)^T$.

$$\begin{aligned}\hat{\tau}_{F_2 * F_3} &= \frac{1}{4}[(4) - (6) - (5) + (9) + (5) - (7) - (8) + (14)] \\ &= \frac{1}{4}[6] = 1.5.\end{aligned}$$

Part (c): Three-factor interaction**Solution**

Using $g_{F_1 * F_2 * F_3} = (-1, +1, +1, -1, +1, -1, -1, +1)^T$ (from part 1c):

$$\begin{aligned}\hat{\tau}_{F_1 * F_2 * F_3} &= \frac{1}{4}[-(4) + (6) + (5) - (9) + (5) - (7) - (8) + (14)] \\ &= \frac{1}{4}[2] = 0.5.\end{aligned}$$

Part (d): Estimated variance and 95% CI for τ_{F_3} **Solution**

With $K = 3$, $N = 16$, and $s_{(w)}^2 = 2$ for all cells:

$$\hat{S}^2 = \frac{1}{2^3} \sum_{w=1}^8 s_{(w)}^2 = \frac{1}{8}(8 \times 2) = 2.$$

$$\widehat{\text{Var}}(\hat{\tau}_F) = \frac{2^K}{N} \hat{S}^2 = \frac{8}{16} \times 2 = 1.$$

So the estimated standard error is $\sqrt{1} = 1$.

A 95% confidence interval for τ_{F_3} is:

$$\hat{\tau}_{F_3} \pm 1.96 \times 1 = 3.5 \pm 1.96 = (1.54, 5.46).$$

Since the interval does not contain zero, we have evidence that F_3 has a nonzero population-average main effect at the 5% significance level.

Part (e): Interpretation**Solution****Main effects.**

- $\hat{\tau}_{F_1} = 2.5$: Switching F_1 from low to high increases the average response by 2.5 units, averaged over all levels of F_2 and F_3 .
- $\hat{\tau}_{F_2} = 3.5$: Switching F_2 from low to high increases the average response by 3.5 units, averaged over all levels of F_1 and F_3 .
- $\hat{\tau}_{F_3} = 3.5$: Switching F_3 from low to high increases the average response by 3.5 units, averaged over all levels of F_1 and F_2 .

Two-factor interactions.

- $\hat{\tau}_{F_1 * F_2} = 1.5 > 0$: The effect of F_1 is larger when F_2 is at its high level. The two factors have a synergistic (positive) interaction.
- $\hat{\tau}_{F_1 * F_3} = 0.5$: A small positive interaction between F_1 and F_3 ; the effect of F_1 increases slightly at the high level of F_3 .
- $\hat{\tau}_{F_2 * F_3} = 1.5 > 0$: A notable synergistic interaction between F_2 and F_3 ; the combined effect of both at their high levels is larger than the sum of individual effects.

The three-factor interaction $\hat{\tau}_{F_1 * F_2 * F_3} = 0.5$ is small, suggesting that the two-factor interactions do not change much depending on the level of the third factor. However, formal inference would be needed to assess whether any of these estimates are statistically distinguishable from zero.

Exercise 3: Fisherian Inference

Part (a): Sharp null hypothesis

Solution

For a 2^3 design, the sharp null hypothesis is: $H_0 : Y_i(-1, -1, -1) = Y_i(-1, -1, +1) = Y_i(-1, +1, -1) = Y_i(-1, +1, +1) = Y_i(+1, -1, -1) = Y_i(+1, -1, +1) = Y_i(+1, +1, -1) = Y_i(+1, +1, +1)$ for *all* units $i = 1, \dots, N$.

That is, every unit's response is completely unaffected by the treatment received — no unit is affected by any factor or any combination of factors.

Part (b): Generating the null randomization distribution of $\hat{\tau}_{F_1}$

Solution

1. Under H_0 , treat each unit's observed outcome Y_i^{obs} as its potential outcome under all eight treatment combinations.
2. Enumerate (or randomly sample) all valid assignments $\mathbf{W}' \in \mathbb{W}^+$, that is, all assignments that allocate exactly $N/8$ units to each of the eight treatment combinations.
3. For each $\mathbf{W}'$, use the re-assigned outcomes to recompute the group means $\bar{Y}_{(w)}^{obs'}$ for each cell w , and then compute $\hat{\tau}'_{F_1} = \frac{1}{4} g_{F_1}^T \bar{Y}^{obs'}$.
4. The null randomization distribution of $\hat{\tau}_{F_1}$ is the collection of all such $\hat{\tau}'_{F_1}$ values. The two-sided p-value is the fraction of these values with $|\hat{\tau}'_{F_1}| \geq |\hat{\tau}_{F_1}^{obs}|$.

In practice (e.g., in R), one would simulate a large number (say $B = 10,000$) of random permutations rather than enumerate all $\frac{N!}{(N/8)!^8}$ assignments.

Part (c): Danger of ignoring interactions

Solution

Focusing solely on main effects can be misleading because a **significant interaction** can either mask or amplify a main effect depending on the chosen level of the other factors.

For example, suppose F_1 has no overall main effect ($\hat{\tau}_{F_1} \approx 0$) but there is a strong positive $F_1 * F_2$ interaction: F_1 has a large positive effect when $F_2 = +1$ and a large negative effect when $F_2 = -1$, and these cancel out in the average. Reporting only $\hat{\tau}_{F_1} \approx 0$ would suggest F_1 is ineffective, when in fact it is highly effective under the right conditions. Interaction effects are often the most actionable finding in an experiment, and ignoring them leads to scientifically incomplete conclusions.

Exercise 4: Extension - Generalization

Part (a): Counting effects

Solution

In a 2^K design, the total number of estimable effects is $2^K - 1$, broken down as:

- K main effects.
- $\binom{K}{2}$ two-factor interactions.
- $\binom{K}{j}$ j -factor interactions, for $j = 2, 3, \dots, K$.

The total is $\sum_{j=1}^K \binom{K}{j} = 2^K - 1$ (by the binomial theorem, since $\sum_{j=0}^K \binom{K}{j} = 2^K$ and the $j = 0$ term is 1).

The number of two-factor interactions specifically is $\binom{K}{2} = \frac{K(K-1)}{2}$.

Part (b): General formula for τ_{i,F_1} and the normalizing constant

Solution

In a 2^K design:

$$\tau_{i,F_1} = \frac{1}{2^{K-1}} Y_i g_{F_1}.$$

The contrast vector g_{F_1} has -1 in the first 2^{K-1} entries (where $F_1 = -1$) and $+1$ in the last 2^{K-1} entries (where $F_1 = +1$), so:

$$\tau_{i,F_1} = \frac{1}{2^{K-1}} \left[\sum_{w: F_1(w)=+1} Y_i(w) - \sum_{w: F_1(w)=-1} Y_i(w) \right].$$

There are 2^{K-1} cells with $F_1 = +1$ and 2^{K-1} cells with $F_1 = -1$. Dividing by 2^{K-1} converts the sums into *averages* over all settings of the other $K - 1$ factors. If we divided by 2^K instead, we would get half the average difference, which would not correspond to the standard definition of a unit-level causal effect.

Part (c): Standard error in a 2^4 design**Solution**

With $K = 4$, $N = 32$, and $\hat{S}^2 = 3$:

$$\widehat{\text{Var}}(\hat{\tau}_F) = \frac{2^K}{N} \hat{S}^2 = \frac{16}{32} \times 3 = \frac{3}{2} = 1.5.$$

The estimated standard error is $\sqrt{1.5} \approx 1.22$ for any effect F in the design.